Lab 45: Prove the template before 35,000 transcripts

Topic 4: Prompt Engineering & Structured Output · Task 4.5: Design efficient batch processing strategies
Runs on Simulator 1: Customer Support Resolution Agent · Run in class
Source: SkillCertPro practice exam Part 1, Q1 (unofficial third-party question)

Commit to an answer first

Before enabling overnight processing of 35,000 archived support transcripts through the Message Batches API, a pilot of 30 transcripts shows inconsistent disposition decisions and missing rationale fields in some outputs. A product manager proposes running the full archive now and resubmitting bad rows the next day. What should this team do next?

A. Submit the full archive as one batch, review the ended results, and resubmit rows whose disposition or fields are wrong.

B. Run the prompt improver once on the current template, accept the revised prompt, and start the full overnight batch.

C. Create a representative evaluation set, refine the templated prompt against clear criteria, and rerun it before submitting the full batch.

D. Add stricter JSON output constraints, keep the current instructions, and use the batch results to discover decision errors.

Simulate the solution

```
# In the browser: open Simulator 1, then Lab 45
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 45

# In a terminal, from simulators/simulator-1-customer-support/
uv run triage.py --pilot --template prompts/triage_template_v0.md
uv run triage.py --pilot
```

What you should see

Weak template: not ready. Real template: ready for the overnight batch.

evidence/triage-triage_template_v0.json has a lower first-pass rate than evidence/triage-triage_template.json; only the latter reports ready_for_batch true.

Open `simulators/simulator-1-customer-support/evidence/` and find it yourself.

The answer

C. Evaluation set, refined template, rerun, then batch.

For a large asynchronous workload the goal is to maximise first-pass success before scale. The video frames prompt engineering as an evaluation loop: define success criteria, build tests that mirror real inputs (routine returns, billing disputes, explicit requests for a human, policy edge cases, messy customer language), iterate on the prompt, validate, then ship. Prompt templates separate fixed instructions from variables such as transcript , customer_history and policy_excerpt , and Console evaluations let you compare prompt versions across test cases. Batch requests are processed independently and asynchronously, so a batch is a poor place to discover preventable prompt defects at full volume. Structured outputs help format reliability, but semantic correctness still needs evaluation.

Why the others fail

A. Uses the 35,000-item batch as the test harness. The pilot already shows systematic defects, so most of the rework is preventable; resubmitting failures by custom_id is for residual errors, not known prompt problems.

B. One unvalidated pass of the prompt improver, accepted without testing against criteria, gives no evidence that the disposition inconsistency is fixed.

D. JSON constraints may fix the missing fields (format), but not the inconsistent decisions (semantics), and errors are still discovered only after the full run.

Check yourself

- the v1 template fills {{transcript}}, {{transcript_id}} and {{policy_excerpt}}
- first-pass success with v1 is 100% on the ten pilot transcripts
- the learner can state why resubmitting bad rows tomorrow costs more than piloting

Now change one thing

Copy a file in cases/ to a new CS id, change the customer's message and expected disposition, rerun with --case, then score it with evaluate.py.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.